

The Coordinate Grid

Use the cue, then fade the cue. Study the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

CUE FOCUS A **coordinate grid** has a sideways **x-axis** and an up-and-down **y-axis** that meet at $(0, 0)$, the **origin**. In the **first quadrant**, both numbers are positive. An **ordered pair** is written (x, y) . To plot a point, always go **over on the x-axis first**, then **up on the y-axis**. For $(3, 2)$, start at the origin, move over 3, then up 2. Order matters: $(3, 2)$ is not the same spot as $(2, 3)$.

Round 1 — Retrieve It

1. What is the sideways axis called?

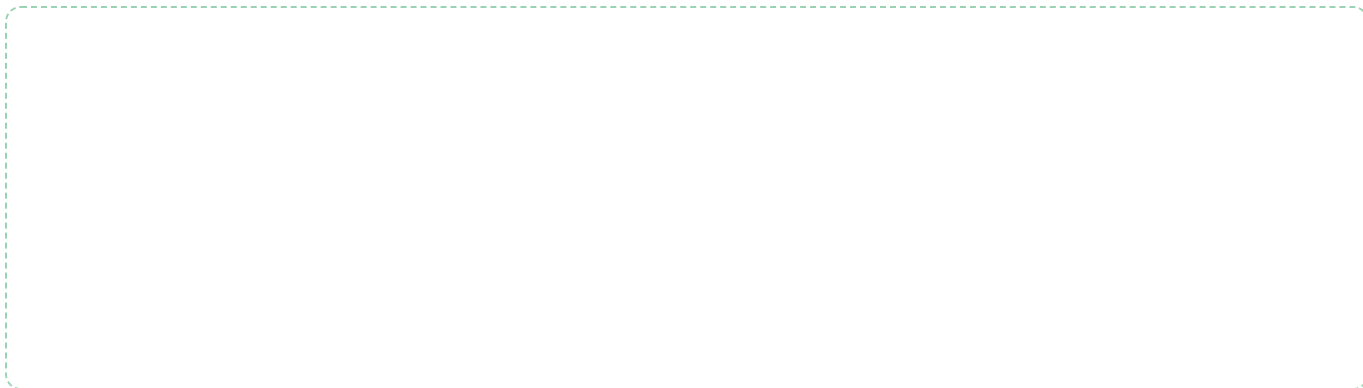
2. In the ordered pair (x, y) , which number tells you how far to go over?

3. When you plot a point, which do you do first, over or up?

4. What point is at $(0, 0)$ called?

Round 2 — Sketch It

Draw a first-quadrant grid. Label the x-axis and y-axis. Plot and label the point $(3, 2)$.



The Coordinate Grid — Apply It

Keep the cue card covered. Answer first. Then use the cue to check, revise, and explain what you changed.

Round 3 — Apply the Idea

1. Tell how to plot (4, 1). Describe each move from the origin.

2. A point is 2 over and 5 up from the origin. Write it as an ordered pair.

3. Explain why (5, 1) and (1, 5) land on different spots.

Round 4 — Reason from Evidence

A friend plotted (3, 2) by going up 3 and over 2. Explain the mistake and tell the correct way to plot (3, 2).

The Coordinate Grid — ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain the steps to plot an ordered pair (x, y) in the first quadrant. Use the words over and up.

C**Connect**

Connect the coordinate grid to a real map or a game with squares. Tell how you would find a spot.

E**Extend**

Choose three ordered pairs that would draw a small shape when connected. List the pairs and name the shape.

Confidence check. Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it.

Teacher Key — The Coordinate Grid

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. The x-axis.
2. The first number, x.
3. Over first (on the x-axis), then up.
4. The origin.

Sketch It — look for

A first-quadrant grid with labeled x-axis and y-axis, and a point plotted at 3 over and 2 up, labeled (3, 2).

Apply It

1. Start at the origin, move over 4 on the x-axis, then up 1 on the y-axis.
2. (2, 5).
3. Order matters: (5, 1) goes over 5 and up 1, while (1, 5) goes over 1 and up 5, so they land in different spots.

Reason from Evidence — look for

The friend switched the order. You must go over on the x-axis first, then up. For (3, 2): over 3, then up 2 (not up 3, over 2).

ACE — Extend look-for

Three first-quadrant pairs that connect into a shape, such as (1, 1), (4, 1), and (1, 4) forming a triangle. Any valid shape with matching pairs is fine.

Suggested use. Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color.